SNES COLLEGE OF ARTS COMMERCE AND MANAGEMENT

(Affiliated to University of Calicut)

PROJECT REPORT

Entitled on

"DRIVER VIGILANCE"

Submitted by:
JISHNU EC (NXAWBCA036)

DEPARTMENT OF COMPUTER SCIENCE

2022-2025

SNES COLLEGE OF ARTS COMMERCE AND MANAGEMENT

DEPARTMENT OF COMPUTER SCIENCE

PROJECT REPORT

"Driver Vigilance Platform for Enhanced Safety and Risk Reduction"

ENTITLED

"DRIVER VIGILANCE"

Under the guidance of

Ms. Aswathi K A

SNES COLLEGE OF ARTS COMMERCE AND MANAGEMENT

CERTIFICATE

This is to certify that **JISHNU EC** is a student of **BACHELOR OF COMPUTER APPLICATION** in S N E S COLLEGE OF ARTS COMMERCE AND MANAGEMENT and the project report entitled on **“DRIVER VIGILANCE”** hasbeen submitted by them on partial fulfillment of the requirements for the award of **BACHELOR OF COMPUTER APPLICATION** degree of the Calicut University during the academic year 2022– 2025.

Place: CHETHUKADAVU

Head of Department of Computer Science

Valued On: …………………..

SNES COLLEGE OF ARTS COMMERCE AND MANAGEMENT

CERTIFICATE

This is to certify that the project work entitled "**DRIVER VIGILANCE**" is a bona fide record of work done by **JISHNU EC** submitted on partial fulfillment of the requirements for the award of degree of **BACHELOR OF COMPUTER APPLICATION** of the Calicut University under my supervision during the academic year 2022 – 2025.

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External viva-voce conducted on____

INTERNAL EXAMINER

EXTERNAL EXAMINER

GET INSPIRED
RISS technologies
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Date: 12/03/2025

CERTIFICATE

This is to certify that **JISHNU E C (Reg. No: NXAWBCA036)** student of **SNES COLLEGE OF ARTS COMMERCE AND MANAGEMENT** has successfully completed his academic project entitled **"DRIVER VIGILANCE"** in **PYTHON, JAVA AND FLUTTER** under the guidance of our senior developers during the period **NOVEMBER 2024 to FEBRUARY 2025.**

During this period he was found hardworking, punctual & efficient. We wish him a successful future.

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DECLARATION

I am **JISHNU EC** hereby declare that the project entitled **"DRIVER VIGILANCE"** submitted to the Calicut University in partial fulfillment of the requirement for the award of degree of **BACHELOR OF COMPUTER APPLICATION** is a record of original work done by me during our period of study at **SNES COLLEGE OF ARTS COMMERCE AND MANAGEMENT** under the supervision and guidance of Ms. Aswathi K A, Head of Computer Science during the academic year 2022-25.

Place: CHETHUKADAVU

Signature of Candidates

Date:__________

JISHNU EC

ACKNOWLEDGEMENT

The success of the project depends upon the effort invested. At this pleasure moment of having successfully completed our project. It is our duty to acknowledge and thanks the individuals who have contributed in the successful completion of the project.

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ABSTRACT

The Driver Vigilance system is a cutting-edge driver monitoring solution designed to detect and prevent distracted driving and drowsiness-related accidents. This innovative system utilizes advanced machine learning algorithms to analyze video inputs of the driver, identifying potential accident-causing behaviors such as phone usage, turning around, or picking items from the rear seat.

The system's primary objective is to enhance road safety by providing real-time alerts and warnings to drivers, thereby minimizing the risk of human error, and ensuring timely assistance. The Driver Vigilance system is designed to integrate advanced technology with practical driving operations, aiming to revolutionize the way drivers stay safe on the road.

By leveraging the power of artificial intelligence and machine learning, the Driver Vigilance system can significantly reduce the number of accidents caused by distracted driving and drowsiness, ultimately saving lives and promoting a safer driving environment. The system is designed to be user-friendly, non-intrusive, and adaptable to various driving scenarios, making it an indispensable tool for drivers, fleet operators, and automotive manufacturers.

The implementation of the Driver Vigilance system has far-reaching implications for road safety, setting a new standard for driver safety and accident prevention. With its real-time monitoring and alert capabilities, this system has the potential to transform the driving experience, providing a safer and more reliable way to navigate roads.

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1. INTRODUCTION

1.1 Problem Statement:

Distracted driving and drowsiness are leading causes of road accidents, resulting in significant loss of life and property. Despite growing awareness about the dangers of distracted driving, the number of accidents attributed to driver inattention and fatigue continues to rise. Traditional methods of preventing distracted driving, such as driver education programs and legislation, have proven insufficient in addressing this critical issue. The lack of real-time monitoring and alert systems for drivers exacerbates the problem, often leading to delayed reactions and increased risk of accidents. This project aims to address these challenges by developing an advanced Driver Vigilance system that utilizes real-time driver monitoring and alert technology to detect and prevent distracted driving and drowsiness-related accidents.

1.2 Working of the Project:

Implementing the Driver Vigilance system involves a combination of computer vision, machine learning, and real-time monitoring technologies to detect and prevent distracted driving and drowsiness-related accidents.

High-level overview of how the system operates:
The Driver Vigilance system utilizes a camera module to capture real-time video footage of the driver. This footage is then processed using advanced machine learning algorithms, which analyze the driver's behavior, facial expressions, and eye movements to detect signs of distraction or drowsiness. Upon detecting any suspicious activity, the system triggers an alert, warning the driver to refocus their attention on the road. The system's real-time monitoring and alert capabilities enable prompt intervention, reducing the risk of accidents caused by distracted driving and drowsiness.

1.3 The system will be able to:

- Implement a driver vigilance system that detects drowsiness and distractions through real-time video analysis.
- Develop a mobile-based driver application with features like registration, login, partner settings, and real-time location sharing.
- Integrate functionalities such as driving pattern detection, voice assistance, and GPS-based road condition notifications.
- Utilize camera assistance to monitor eye closures and detect drowsiness, providing timely alerts to the driver.
- Detect driver distractions like phone usage or other non-driving activities and issue alerts to maintain focus on the road.
- Synchronize data and services between driver, partner, and emergency response apps (ambulance, hospital, and police).
- Provide emergency help functionalities that allow drivers to directly contact police, hospitals, or ambulance services in case of accidents.
- Enable partner and website modules for system synchronization, along with feedback mechanisms for continuous improvement of the system.

Advantages:

- Enhanced Road Safety: Early detection of drowsiness and distractions helps prevent accidents effectively.
- Comprehensive Emergency Support: Direct connections to ambulance, hospital, and police ensure quick responses in emergencies.
- User-Friendly Interface: Intuitive app interfaces support easy navigation and usability for drivers and partners.
- Increased Driver Awareness: Notifications about road conditions and driving patterns enhance driver awareness and preparedness.
- Technology Integration: Combines AI, GPS, and mobile sensors for advanced vigilance and safety monitoring.
- System Synchronization: Seamless integration between driver, partner, and emergency apps ensures efficient communication and response.

Limitations/Disadvantages:

- Requires a stable internet connection for optimal performance.
- Users must register and provide personal details, which could pose data privacy concerns if mishandled.
- Effective vigilance relies on the camera's image quality, which may vary under different lighting conditions.
- Hardware requirements like smartphones and cameras with sufficient processing capability are essential.

System Description:

The system comprises three major modules

1. Driver App:

- Registration and Login
- Partner Settings and Auto Messaging
- Location Sharing and Driving Pattern Detection
- Voice and Camera Assistance
- Emergency Support for Police, Ambulance, and Hospital

2. Website Module:

- User account management
- System synchronization with mobile apps
- Feedback and complaint management
- Emergency and hospital-related data sharing

3. Emergency Apps (Ambulance, Hospital, Police):

- Direct communication channels for emergency situations
- Access to driver location and alert details
- Support for accident response and coordination.

2. SYSTEM ANALYSIS

System analysis is a general term that refers to an orderly, structured process for identifying and solving a problem. The system analysis process is called the life cycle methodology, since it relates to four significant phases in the life cycle of all business information system: study, design, development and operation. The definition of system analysis includes not only the process but also the process of putting together to form a new system. A system analyst is an individual who performs system analysis during any or all of the life cycle phases of a business information system. The system analyst not only analyses business information system problems, but also synthesizes new systems to solve those problems or meet other information needs. The various techniques used in the study of the present system are:

- Observation
- Interviews
- Site visits
- Discussion

Preliminary Investigation

Preliminary investigation checks whether a system is developed by means of SDLC, a prototyping strategy or structured analysis method or combination of these methods. A project request should first be reviewed. The choice of the development strategy of the project is secondary to investment of an organization resource in information system project. The entire proposal for the required project is submitted to the selection committee for evaluation to identify that these projects are most beneficial to the organization. The preliminary investigation is thus carried out by the system analyst under the direction of the selection committee. In this stage at first visited an office to know

Identification of needs

The first step in the SDLC is the identification of needs. Since there is likely to be stream of user's requests, standard procedures must be established to deal with them, the initial investigation is one way of handling this. The objective is to determine whether the request is valid and feasible before an improvement or modification of an existing or building a new system

Fact Finding Techniques

There are several methods for gathering the sort of information. We can use all of these methods for gathering information from the user of the existing system. We can introduce seven fact-finding techniques.

Sampling

Sampling is the process of collecting a representative sample of documents, forms and records. Because it would be impractical to study every occurrence of every form or record in a file or database, system analyst normally uses sampling techniques to get a large enough cross section to determine what can happen in the system. The system analyst seeks to sample enough forms to represent the full nature and complexity of the data. First collected sample is a Sample receipt and conducted a study to know how these data can be converted to a digital method

Research and Visit Sites

Another fact- finding technique is to thoroughly research the problem domains. Most problems are not unique. Others have solved them before us. We can contact or perform site visits at companies that have experienced similar problems. If these companies are willing to share, valuable information can be obtained, may be tremendous time and cost in the development process. Computer trade journals and reference books are also a good source of information from the collected receipt we found how collection details are recorded in a receipt. The next point is how it is recorded in an office register. We must design a database such that all that data must be maintained in that database.

Observation

Observation is fact finding technique where in the system analyst either participates in or watches a person perform activities to learn about the system. This technique is often used when the validity of data collected through other methods is in question or when the complexity of certain aspects of the system prevents a clear explanation by the end users. This is an effective data- collection technique for obtaining an understanding of a system

Interviews

These are fact finding techniques where by the system analyst Collects information from individuals through face-to-face interaction. The personal interview is generally recognized as the most important and most often used fact finding techniques. Interviewing can be used to achieve any or all of the following goals: find facts, verify facts, clarify facts, generate enthusiasm, get the end-user involved, identify requirements and solicit ideas and opinions.

Cost Benefit Analysis

“Cost Benefit Analysis (CBA) estimates and totals up the equivalent money value of the benefits and cost to the community of projects to establish they are worthwhile. “In order to reach a conclusion as to the desirability of a project, all aspects of the project, positive and negative, must be expressed in terms of a common unit; i.e. There must be a “bottom line”. The most convenient common unit is money. A program may provide benefits which are not directly expressed in terms of dollars but there is some amount of money the recipients of the benefits would consider just as the project’s benefits. When all data have been identified and broken down into cost categories, the analyst must select a method for evaluation. Several evaluation methods are available, each with pros and cons

The common methods are:

i. Net benefit analysis
ii. Present value analysis
iii. Net present value
iv. Payback analysis
v. Break even analysis
vi. Cash flow analysis

After completing all these phases, a simple idea has been generated such that we can design a system that can limit the drawbacks of existing system. The next point to be focused is its merit and demerit. One of the important facts that the limit the performance is that a position that can cover main route of goods must be chosen. Integration of map application will arrange the felicity to do this.

2.1 EXISTING SYSTEM

Existing systems to address distracted and drowsy driving primarily involve technological solutions, public awareness campaigns, and legislative measures. Technological solutions include in-car monitoring systems, such as lane departure warnings, collision avoidance systems, and driver attention detection systems that alert drivers when signs of distraction or drowsiness are detected. Smartphone applications can also disable certain functions while driving to reduce distractions. Public awareness campaigns aim to educate drivers about the dangers of distracted and drowsy driving. Meanwhile, stricter laws and enforcement efforts, such as penalties for using mobile phones while driving, are being implemented in many regions to deter these behaviours.

Disadvantages of the Existing System

- **Limited Effectiveness of Technological Solutions** – While in-car monitoring systems and smartphone applications provide alerts, they often fail to address the root causes of distraction or drowsiness. False alarms or missed detections may also reduce driver trust in these systems.
- **Dependency on User Compliance** – Smartphone applications designed to reduce distractions rely heavily on drivers enabling and adhering to these tools. Non-compliance by users can render such solutions ineffective.
- **High Cost of Implementation** – Advanced technological systems like driver attention detection and collision avoidance require significant investment, making them unaffordable for many vehicle owners.
- **Public Awareness Campaign Limitations** – While awareness campaigns educate drivers, they cannot ensure consistent behavioral change. The impact of such campaigns may diminish over time without continuous reinforcement.
- **Enforcement Challenges** – Stricter laws and penalties are difficult to enforce uniformly, especially in regions with limited resources. Some drivers may still find ways to bypass regulations.
- **Reactive Nature** – Most existing systems and measures are reactive, addressing the issue after it occurs, rather than proactively preventing distracted or drowsy driving

2.2 PROPOSED SYSTEM

The proposed system is divided into three parts,

1. Driver App
2. Partner App
3. Website for synchronising both app
4. Camera based driver vigilance control
5. Ambulance App
6. Hospital App
7. Police App

Functions of Driver App

1. Registration
2. Login
3. Partner Setting
4. Auto messaging to own contacts
5. Location Sharing
6. Driving pattern detection using mobile sensors
7. Voice assistance support
8. Camera assistance while closing eye lids(sleeping)
9. Get road condition notifications using GPS, accelerometer
10. View contact book
11. Sent emergency help to police , ambulance, hospital
12. View hospital
13. View doctors
14. View ambulance details

Functions of partner app

1. Login
2. View drivers (partners)
3. Driver phone mode settings
    a. Speed based
    b. Vehicle angle based

   Main settings are,

   1. Call blocking
   2. Screen light intensity setting
   3. Screen lock while speed increases
   4. Auto smsing service
   5. Ring mode settings
4. View Driving logs
5. Messaging to partner
6. View Locations

Function of ambulance app

1. Signup
2. Login
3. View profile
4. Location updation
5. Working status updation
6. View emergency help from users
7. Track emergency help requested person using google map

Functions of hospital app

1. Signup
2. Login
3. View profile
4. Doctors management
5. View help from users

Functions of police

1. Signup
2. Login
3. View profile
4. Contact book management
5. View help from users and track
6. Sent location based notification

Functions of Website:

1. Controls and manage the data flow between partner and driver

2.3 FEASIBILITY STUDY

A feasibility study evaluates the practicality of implementing the Driver Vigilance System, considering technical, economic, legal, operational, behavioral, software, and hardware viability. This study helps in identifying potential challenges and benefits before investing significant time and resources into the project.

1.Technical Feasibility:

- The project utilizes proven technologies like cameras, image processing, and machine learning techniques to detect driver distractions. These are readily available and widely supported.
- Python (Flask) as the front-end framework and MySQL as the database make it a robust solution. Both are free and have extensive community support.
- Cloud storage and GPS tracking enable real-time updates, ensuring efficient alert systems for drivers and external parties like ambulances or police.
- The project is technically feasible since its components align with existing, reliable, and scalable technologies.

2. Economic Feasibility:

- Initial costs might include investments in camera hardware, cloud infrastructure, and development resources. However, the long-term benefits, such as reduced accidents, enhanced road safety, and time-saving alerts, outweigh these expenses.
- Collaboration with government road safety initiatives or transportation agencies could potentially reduce development costs and garner funding.
- This system is cost-effective, as it integrates multiple functionalities, such as location sharing, notification systems, and partner apps, into a single platform.

3. Legal Feasibility:

- The system must comply with traffic and road safety regulations, especially when integrating features like GPS tracking and emergency notifications.
- Data privacy laws need to be adhered to, ensuring that driver data (such as real-time location and camera footage) is securely stored and not misused.
- Complying with global and regional laws for emergency response services (police, hospitals, ambulances) will ensure a smooth rollout.

4. Operational Feasibility:

- The system enhances road safety by integrating AI-driven vigilance monitoring with a user-friendly app for drivers and their partners.
- Minimal training is required for drivers, partners, and emergency services to use the platform effectively.
- Its scalability supports adoption across diverse geographies and integration with existing road safety measures.
- Features like automatic messaging, voice alerts, and distraction detection make it highly practical for real-world use.

Technical Feasibility:

The technical requirements for the system are economical and do not involve the need for additional or proprietary software. The application is developed using Python, a widely-used programming language with free, readily available development tools. The use of frameworks like Flask and cloud-based services ensures real-time processing and system scalability. Additionally, technologies like MySQL for database management and accessible hardware (like cameras and smartphones) make the system technically feasible and practical to implement.

Economical Feasibility

The economic analysis evaluates the benefits and savings expected from the proposed system compared to its costs. Although there might be higher initial investments due to the purchase of camera devices and development costs, the system proves to be cost-effective and time-efficient in the long run. The tangible benefits include fewer accidents and a reduction in emergency response delays, while intangible benefits include improved road safety and user convenience. By adopting cost-benefit analysis techniques, it is evident that the system is economically feasible as the quality of service it provides justifies the cost.

Operational Feasibility

The success of the system in an operational environment relies on how effectively it integrates into current practices. The system’s design, with user-friendly features such as voice alerts, distraction detection, and emergency notifications, ensures ease of use. Minimal training for drivers and partners makes it practical to deploy widely. Furthermore, the system enhances the efficiency of police, ambulance, and hospital coordination, ensuring it meets operational requirements seamlessly. Its scalability and adaptability for different regions further validate its operational feasibility.

Behavioral Feasibility

This analysis examines the system's acceptance by end users and stakeholders. While initial resistance to change might be expected, the life-saving potential of the system ensures it is broadly accepted once its benefits are understood. With intuitive interfaces, clear alerts, and improved safety outcomes, users are likely to embrace the system. Educational campaigns and demonstrations can further overcome hesitance, ensuring the new system gains wide acceptance among drivers and emergency responders**.**

Software Feasibility

Even though this application is developed in very high software environment, it is also supportedby many other environments with minimum changes. The system is fully feasible to be executed on any kind of operating systems and browsers.

Hardware Feasibility

Software can be developed with the existing resources. But the existing resources may or may not be used to produce hardware. If no hardware is newly bought for project, then software is said to achieve hardware feasibility. The system is hardware wise feasible because it needed absolutely no new hardware

2.4 SYSTEM SPECIFICATION

The system specification refers to a detailed functional and non-functional description of a system. This term can also be defined as an explicit set of requirements that need to be satisfied be specific system. System specification includes software and hardware specification of project

HARDWARE SPECIFICATION:

The selection of hardware is very important in the existence and proper working of any software. Then selection hardware, the size and capacity requirements are also important.

Processor: Intel Core 2 Dual and above

Android: Version 7 and Above

Primary Memory: 1 GB RAM and above

Storage: 40 GB hard disk and above

Display: VGA Color Monitor

Key Board: Any OS compatible Mouse: Any OS compatible

SOFTWARE SPECIFICATION:

One of the most difficult tasks is selecting software for the system, once the system requirements is found out then we have to determine whether a particular software package fits for those system requirements. The application requirement:

- Front end: HTML/CSS, Android
- Back end: Python
- Database: MySQL
- Operating system: Windows 7 and above

2.5 SOFTWARE TOOLS USED

PyCharm (Python Interpreter)

Python is an interpreter, object-oriented, high-level programming language with dynamic semantics. Its high-level built-in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed. Often, programmers fall in love with Python because of the increased productivity it provides. Since there is no compilation step, the edit-test-debug cycle is incredibly fast. Debugging Python programs is easy: a bug or bad input will never cause a segmentation fault. Instead, when the interpreter discovers an error, it raises an exception. When the program doesn't catch the exception, the interpreter prints a stack trace. A source level debugger allows inspection of local and global variables, evaluation of arbitrary expressions, setting breakpoints, stepping through the code a line at a time, and so on. The debugger is written in Python itself, testifying to Python's introspective power. On the other hand, often the quickest way to debug a program is to add a few print statements to the source: the fast edit-test-debug cycle makes this simple approach very effective. Python is a multi-paradigm programming language. Object-oriented programming and structured programming are fully supported, and many of its features support functional programming and aspect-oriented programming (including by Meta programming and meta objects (magic methods)). Many other paradigms are supported via extensions, including design by contract and logic programming. Python uses dynamic typing, and a combination of reference counting and a cycle-detecting garbage collector for memory management. It also features dynamic name resolution (late binding), which binds method and variable names during program execution

Android

Android is an operating system based on the Linux kernel. The project responsible for developing the Android system is called the Android Open-Source Project (AOSP) and is primarily lead by Google. The Android system supports background processing, provides a rich user interface library, supports 2-D and 3-D graphics using the OpenGL-ES (short OpenGL) standard and grants access to the file system as well as an embedded SQLite database. An Android application typically consists of different visual and non-visual components and can reuse components of other applications. The Android system is a full software stack, which is typically divided into the four areas. The levels can be described as

- Applications - The Android Open-Source Project contains several default applications, like the Browser, Camera, Gallery, Music, Phone and more.
- Application framework - An API which allows high-level interactions with the Android system from Android applications
- Libraries and runtime - The libraries for many common functions (e.g.: graphic rendering, data storage, web browsing, etc.) of the Application Framework and the Dalvik runtime, as well as the core Java libraries for running Android applications

- Linux kernel - Communication layer for the underlying hardware.

The Linux kernel, the libraries and the runtime are encapsulated by the application framework. The Android application developer typically works with the two layers on top to create new Android applications. The Android Software Development Kit (Android SDK) contains the necessary tools to create, compile and package Android applications. Most of these tools are command line based. The primary way to develop Android applications is based on the Java programming language.

"The Android Developer Tools (ADT) were previously based on the Eclipse IDE, but Google now recommends using Android Studio for Android application development. Android Studio is built on IntelliJ IDEA, providing all necessary functionality for creating, compiling, debugging, and deploying Android applications. It also includes tools for managing virtual Android devices for testing purposes.

Both Android Studio and Eclipse/ADT offer specialized editors for Android-specific files, allowing developers to switch between XML representation and structured user interfaces. Eclipse's plug-in architecture, based on Equinox, supports extending the IDE's functionality to work with various programming languages, networking applications, and database management systems.

Java and CVS support is provided in the Eclipse SDK, with additional version control systems supported by third-party plug-ins. The Eclipse SDK includes Eclipse Java Development Tools (JDT), offering advanced refactoring techniques and code analysis. Eclipse utilizes the Standard Widget Toolkit (SWT) for graphical control elements, with JFace providing a UI toolkit and framework simplifying application development based on SWT.

Android applications are developed using the Java programming language. The Android SDK includes the Android Debug Bridge (adb), facilitating communication between development machines and Android devices/emulators for debugging and other purposes."

DREAMWEAVER

Adobe Dreamweaver is a software program designed for creating and editing web pages, offering a comprehensive HTML web and programming editor. The program provides a What-You-See-Is-What-You-Get (WYSIWYG) interface, allowing users to design web pages in a user-friendly environment.

Dreamweaver supports multiple markup languages, including HTML and Extensible Markup Language (XML), as well as style sheet languages like Cascading Style Sheets (CSS). Additionally, it supports various programming languages such as JavaScript, C#, Visual Basic (VB), Active Server Pages (ASP), and others.

The program is available in numerous languages, including English, Spanish, French, German, Japanese, Chinese (both Simplified and Traditional), Italian, Russian, and more. Dreamweaver was originally developed and published by Macromedia in 1997. Adobe acquired Macromedia (and the rights to Dreamweaver) in 2005, continuing the development of the software.

With its many features, Dreamweaver serves as a versatile web editing tool suitable for creating both complex and simple websites.

2.6 DATABASE SERVICES

A database is a collection of interrelated data stored with minimal redundancy to efficiently serve various purposes. The primary objective is to ensure that accessing information is easy, quick, inexpensive, and flexible for users. In database design, several objectives are considered, including controlling redundancy, ease of learning and use, minimizing data dependence, maximizing information retrieval at a low cost, and ensuring accuracy and integrity

3. SYSTEM DESIGN

3.1 INTRODUCTION

The most creative and challenging phase of the system develop mentis the system design. It provides the understanding and procedural details necessary for implementing the system recommended in the feasibility study. Design goes through the logical and physical stages of development. In designing a new system, the analyst must have a clear understanding of the objectives, which the design is aiming to fulfill. The first step is to determine how the output is to be produced and in what format. Second input data and master files have to be designed to meet the requirements of the proposed output. The operational phases are handled through program construction and testing. Finally, details related to justification of the system and an estimate of the impact of the candidate system on the user and the organization are documented a devaluated by the management. Design of a system can be defined as a process of applying various techniques and principles for the purpose of defining a device, a process or a system in sufficient detail to permit its physical realization. Thus, system design is a solution, “how to” approach to the creation of a new system. The design step provides a data design, architectural design and a procedural design. The data design transforms the information domain created during analysis into the data structure that will be required to implement the software. The architectural design defines the relationship among major structural components and procedural description of the software. Source code is generated and testing conducted to integrate and validate the software.

System design goes through two phases of development:

- Logical design
- Physical design

Logical Design

The part of the design process that is independent of any specific hardware or software platform is referred to as logical design. During logical design, all functional features of the system chosen for development in analysis phase are described independently of any computer platform. Logical design concentrates on the business aspects of the system and tends to be oriented to a high level of specificity.

Physical Design

Physical design is the part of the design phase in which the logical specifications of the system from logical design are transferred into technology-specific details from which all programming and system construction can be accomplished. As a part of the physical design, analysts design the various parts of the system to perform the physical operation necessary to facilitate data capture, processing, and information output.

3.2 INPUT DESIGN

The first step in system design is to design input and output within predefined guidelines. In input design, user originated inputs are converted into computer-based format. In output design, the emphasis is on producing the hard copy of the information requested or displaying the output on a CRT screen in a predefined format. The following features have been incorporated into the input design of the proposed system.

Easy Data Input

Data entry has been designed in a manner much like the source documents.

Appropriate messages are provided in the message area, which prompts the user in entering the right data. Erroneous data inputs are checked at the end of each screen entry.

Data Validation

The input data is validated to minimize errors in data entry. For certain data specific codes have been given and validation is done which enables the user to enter the required data or correct them if they entered wrong codes.

User Friendliness

User is never left in a state of confusion as to what is happening, instead appropriate error and acknowledge messages are sent. Error maps are used to indicate the error codes and specific error messages.

Consistent Format

A fixed format is adopted for displaying the title messages. Every screen has line, which displays the operation that can be performed after the data entry. They are normally done at the touch of a key.

Interactive Dialogue

The system engages the user in an interactive dialogue. The system is able to extract missing or omitted information from the user by directing the user through appropriate messages, which are displayed.

OUTPUT DESIGN

The output is the most important and direct source of information to the user. The output should be provided in a most efficient formatted way. Based on the options given by the users and the administrator various types of output screens have been generated. The computer output is the most important and direct source of information to the user. Efficient and intelligible output design improves the system’s relationship with the user and helps in decision- making. Output design was studied going actively during the study phase. The objective of the output design is defined the contents and format of all documents and reports in an attractive and useful format.

3.3 DATA FLOW DIAGRAM

A data flow diagram (DFD) is a graphical representation of the flow of data

through an information system, modeling its process aspects. A DFD shows what kinds of information will be input to and output from the system, where the data will come from and go to, and where the data will be stored. These are expanded by level, each explaining its process in detail. Processes are numbered for easy identification and are normally labeled in block letters. Each Data flow is labeled for easy understanding. Data flow diagrams are made up of a number of symbols, which represents system components. Data flow modeling method uses four kinds of symbols.

Process

Process shows the work of the system. Each process has one or more data inputs and produce

one or more data outputs. Processes are represented by circles in data flow diagrams.

Data Stores

A data store is a repository of data. Processes can enter data into a store or retrieve data from the data store. Data stores are represented by two parallel lines, which may be depicted horizontally or vertically.

Data Flows

The arrows represent data flow. A data flow is data in motion. A data flow represents an input of data to a process or the output of the data from a process. A dataflow is also used to represent the creation, reading, deletion, or updating of data in a file or database.

External Entities

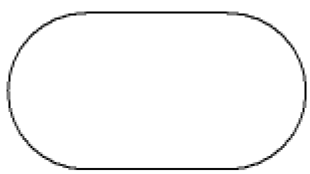
External entities are outside the system but they either supply input the system or use other systems output. They are entities on which the designer has control. External entities that supply data into the system are sometimes called source. External entities that use the system data are called sinks. These are represented by rectangles in the data flow diagram.

In DFD there are four main symbols: -

- A square defines a Source or destination of a system data

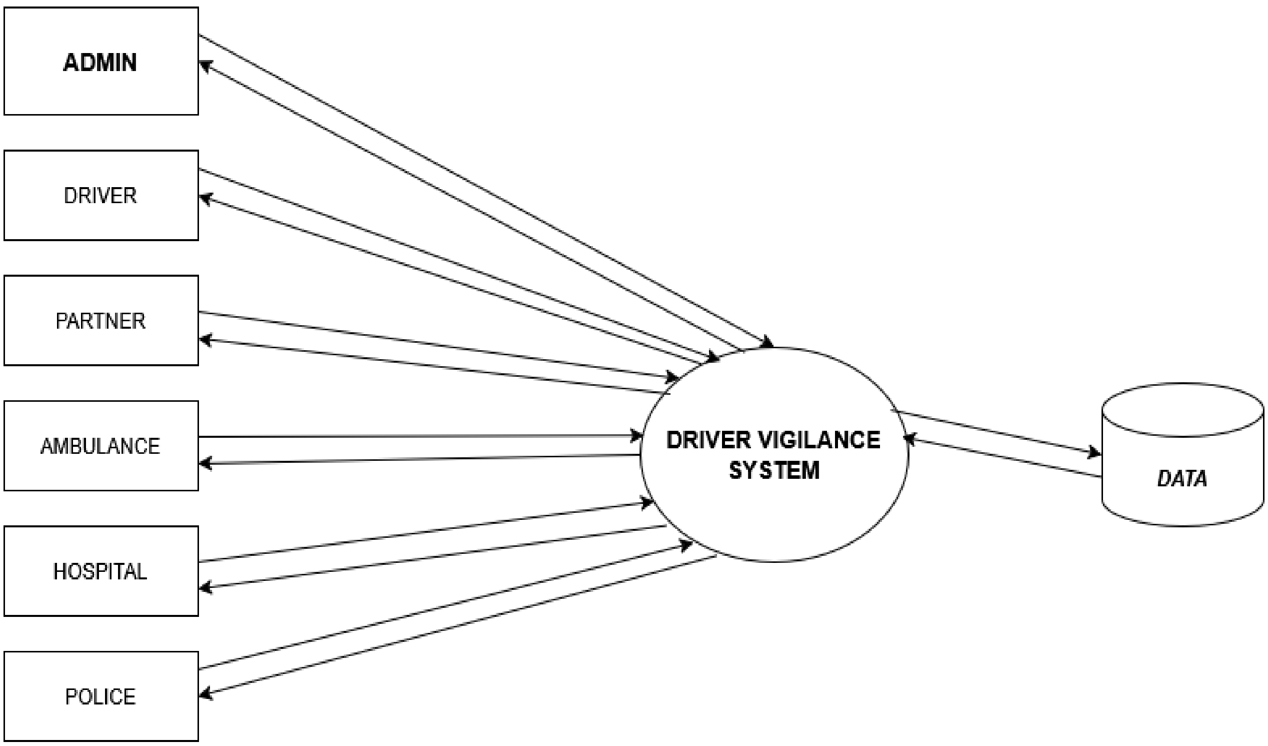
- An arrow identifies data flow. It is a pipeline through which information flows

- A circle represents that transforms incoming data flow into outgoing data flow



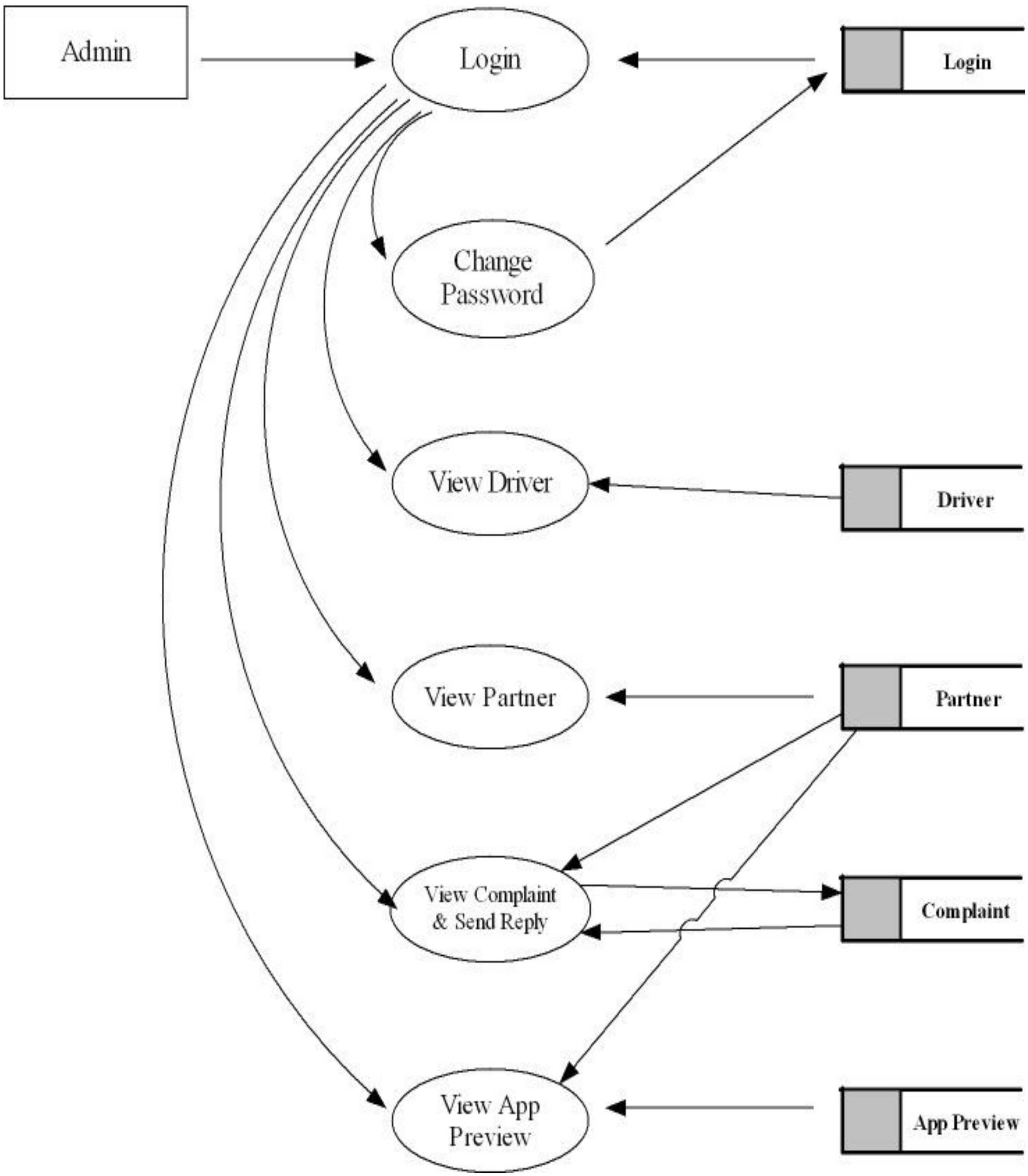
- An open rectangle stores data

Level 0

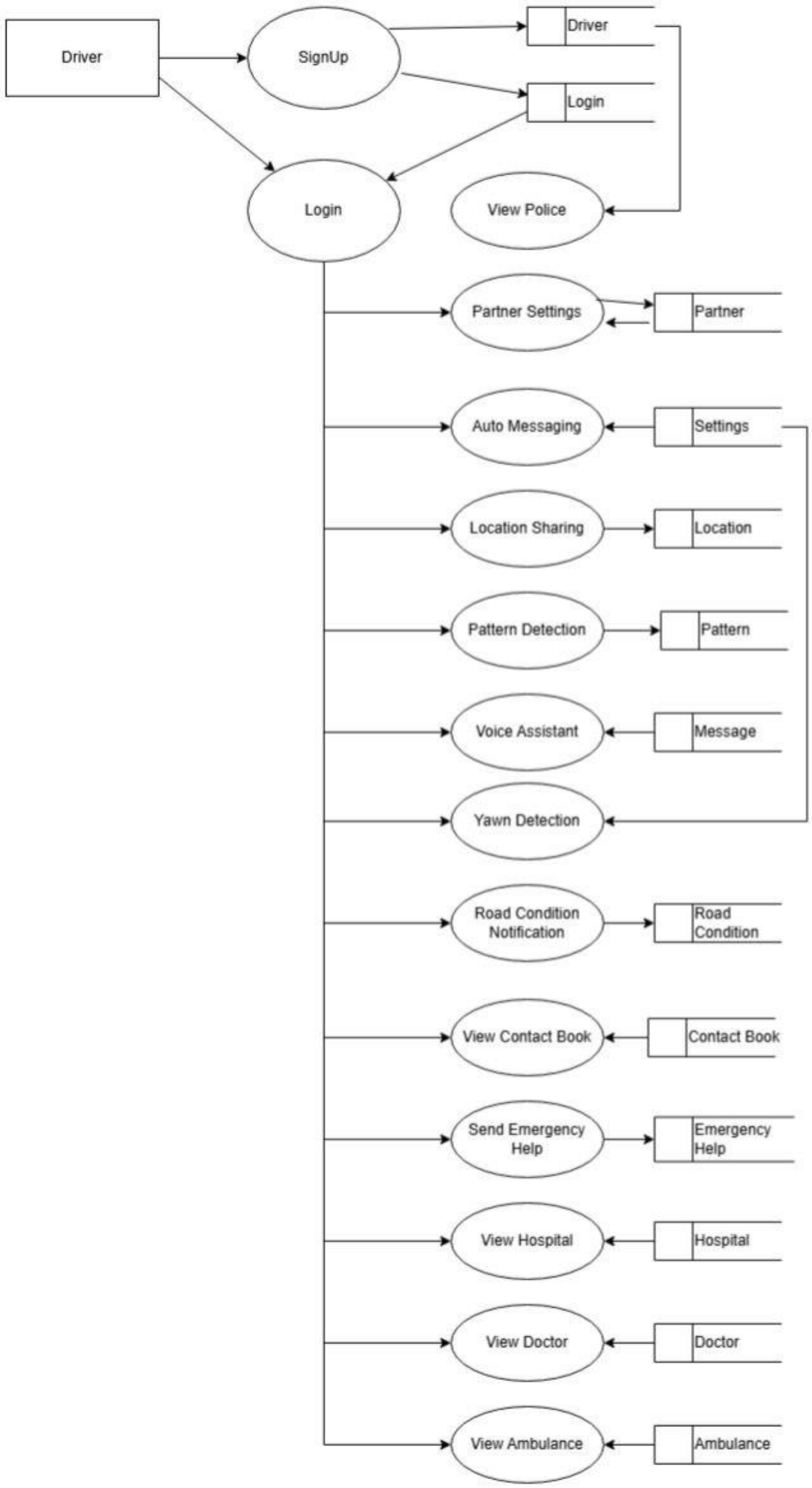


ADMIN
DRIVER
PARTNER
AMBULANCE
HOSPITAL
POLICE
DRIVER VIGILANCE SYSTEM
DATA

Level 1

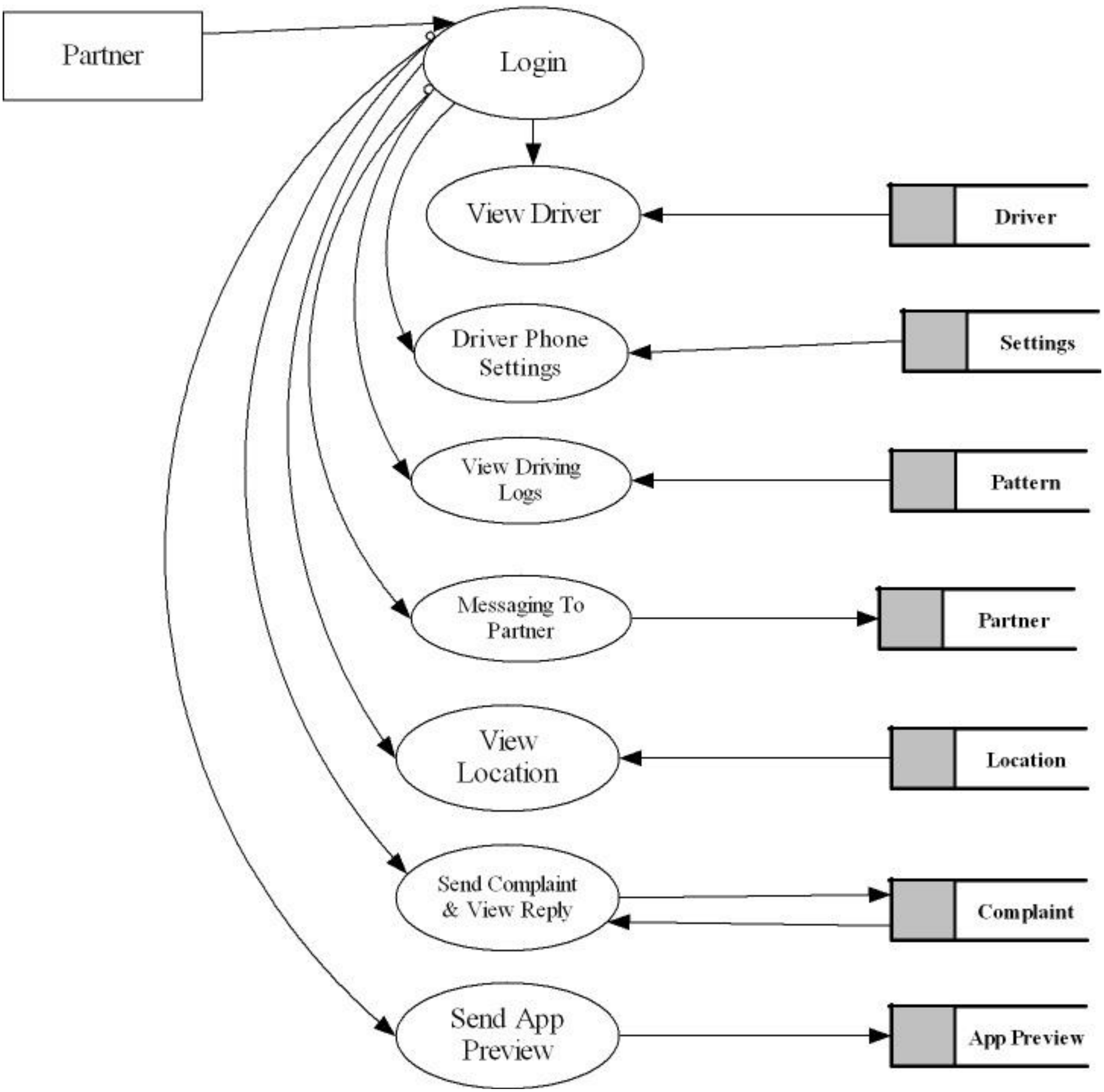

Admin
Login
Login
Change Password
View Driver
Driver
View Partner
Partner
View Complaint & Send Reply
Complaint
View App Preview
App Preview

Level 2



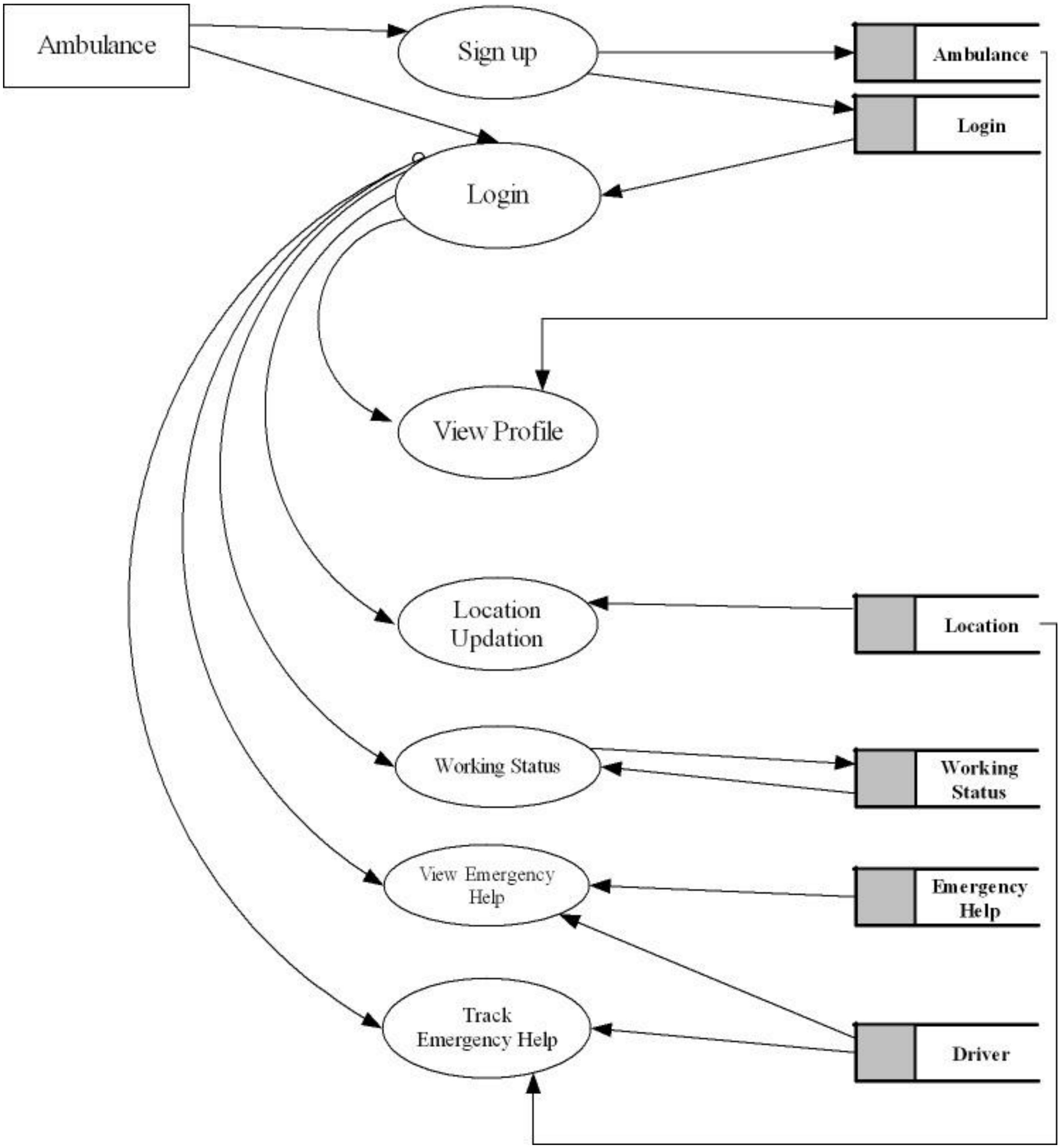
Driver
SignUp
Driver
Login
Login
View Police
Partner Settings
Partner
Auto Messaging
Settings
Location Sharing
Location
Pattern Detection
Pattern
Voice Assistant
Message
Yawn Detection
Road Condition Notification
Road Condition
View Contact Book
Contact Book
Send Emergency Help
Emergency Help
View Hospital
Hospital
View Doctor
Doctor
View Ambulance
Ambulance

Level 3

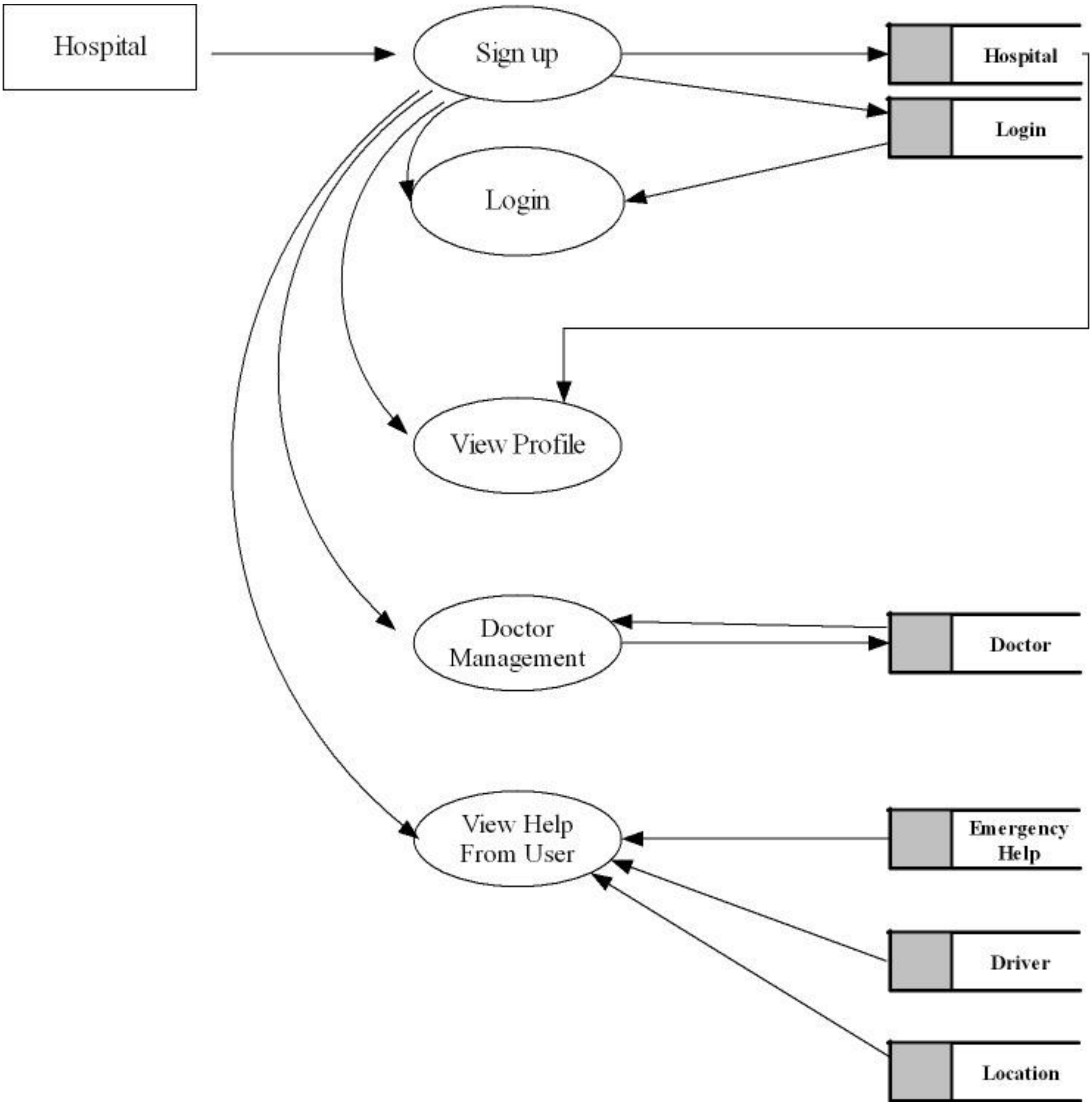


Partner
Login
View Driver
Driver
Driver Phone Settings
Settings
View Driving Logs
Pattern
Messaging To Partner
Partner
View Location
Location
Send Complaint & View Reply
Complaint
Send App Preview
App Preview

Level 4

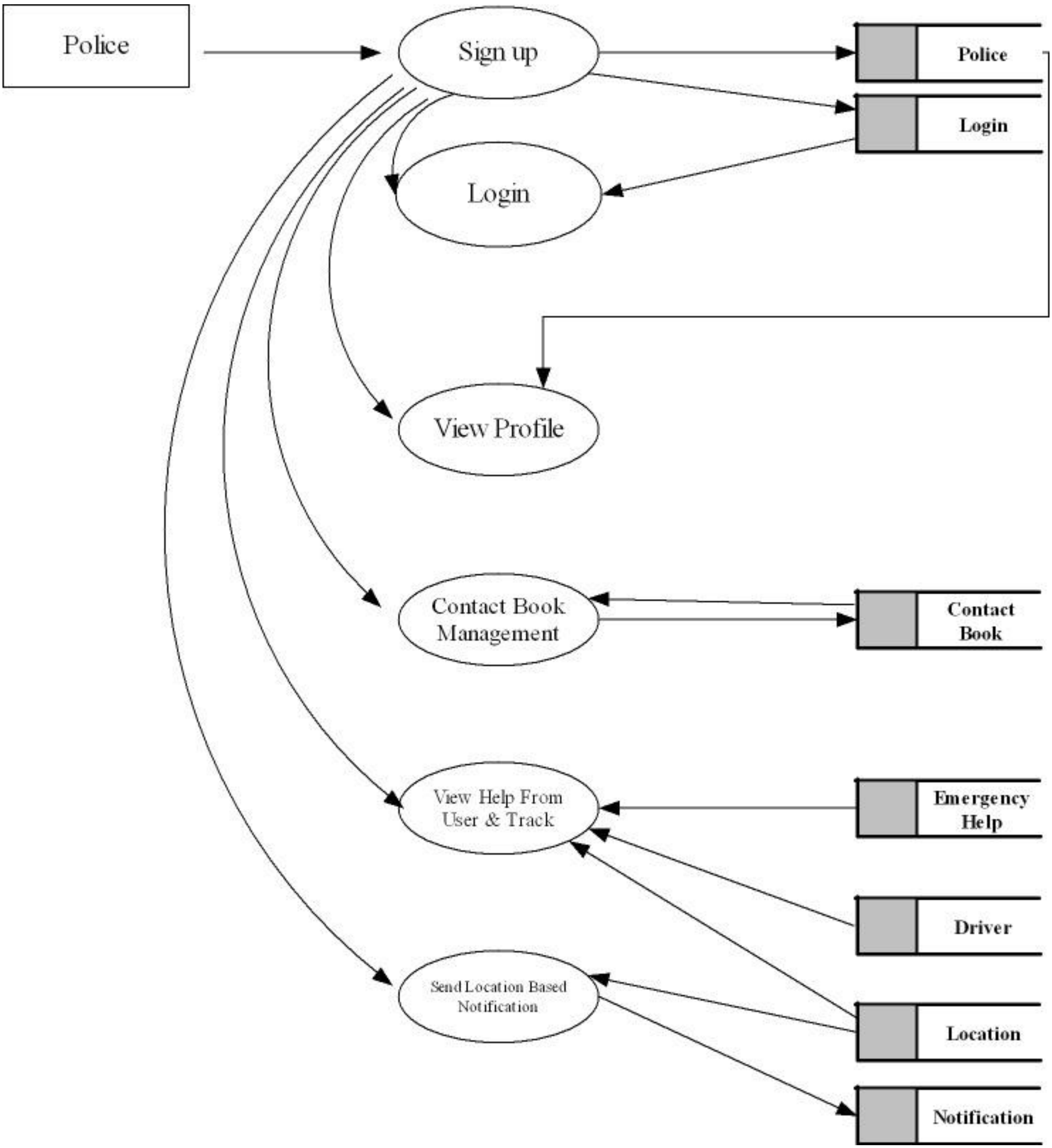

Ambulance
Sign up
Ambulance
Login
Login
View Profile
Location Updation
Location
Working Status
Working Status
View Emergency Help
Emergency Help
Track Emergency Help
Driver

Level 5



Hospital
Sign up
Hospital
Login
Login
View Profile
Doctor Management
Doctor
View Help From User
Emergency Help
Driver
Location

Level 6


Police
Sign up
Police
Login
Login
View Profile
Contact Book Management
Contact Book
View Help From User & Track
Emergency Help
Driver
Send Location Based Notification
Location
Notification

3.4 TABLES

TABLE NAME:LOGIN

	Field	Type	Comment
	id	int(11) NOT NULL	
	username	varchar(100) NOT NULL	
	password	varchar(100) NOT NULL	
	type	varchar(100) NOT NULL	

TABLE NAME: AMBULANCE

	Field	Type	Comment
	id	int(11) NOT NULL	
	drivername	varchar(100) NOT NULL	
	phnno	varchar(100) NOT NULL	
	license	varchar(150) NOT NULL	
	vehicleno	varchar(100) NOT NULL	
	LOGIN_id	int(11) NOT NULL	
	rc_proof	varchar(100) NOT NULL	
	email	varchar(100) NOT NULL	

TABLE NAME: DISRUPTIONS

	Field	Type	Comment
	id	int(11) NOT NULL	
	place	varchar(100) NOT NULL	
	latitudinal	varchar(100) NOT NULL	
	longitudinal	varchar(100) NOT NULL	
	date	date NOT NULL	

TABLE NAME: DOCTOR

	Field	Type	Comment
	id	int(11) NOT NULL	
	name	varchar(100) NOT NULL	
	specialization	varchar(100) NOT NULL	
	phnno	varchar(100) NOT NULL	
	place	varchar(100) NOT NULL	
	pin	varchar(100) NOT NULL	
	post	varchar(100) NOT NULL	
	district	varchar(100) NOT NULL	
	HOSPITAL_id	int(11) NOT NULL	
	id_proof	varchar(150) NOT NULL	

TABLE NAME: DRIVER

	Field	Type	Comment
	id	int(11) NOT NULL	
	name	varchar(100) NOT NULL	
	email	varchar(100) NOT NULL	
	photo	varchar(150) NOT NULL	
	phnno	varchar(100) NOT NULL	
	place	varchar(100) NOT NULL	
	pin	varchar(100) NOT NULL	
	district	varchar(100) NOT NULL	
	post	varchar(100) NOT NULL	
	license	varchar(200) NOT NULL	
	LOGIN_id	int(11) NOT NULL	

TABLE NAME: DRIVER LOG

	Field	Type	Comment
	id	int(11) NOT NULL	
	date	varchar(100) NOT NULL	
	time	varchar(100) NOT NULL	
	speed	varchar(100) NOT NULL	
	angle	varchar(100) NOT NULL	
	latitude	varchar(100) NOT NULL	
	longitude	varchar(100) NOT NULL	
	place	varchar(100) NOT NULL	
	DRIVER_id	int(11) NOT NULL	

TABLE NAME: EMERGENCY

	Field	Type	Comment
🔑	id	int(11) NOT NULL	
	date	varchar(100) NOT NULL	
	emergency	varchar(100) NOT NULL	
	time	varchar(100) NOT NULL	
	DRIVER_id	int(11) NOT NULL	
	latitude	varchar(100) NOT NULL	
	longitude	varchar(100) NOT NULL	

TABLE NAME: FEEDBACK

	Field	Type	Comment
🔑	id	int(11) NOT NULL	
	date	date NOT NULL	
	feedback	varchar(100) NOT NULL	
	LOGIN_id	int(11) NOT NULL	

TABLE NAME: HOSPITAL

	Field	Type	Comment
🔑	id	int(11) NOT NULL	
	name	varchar(100) NOT NULL	
	phnno	varchar(100) NOT NULL	
	email	varchar(100) NOT NULL	
	place	varchar(100) NOT NULL	
	pin	varchar(100) NOT NULL	
	post	varchar(100) NOT NULL	
	district	varchar(100) NOT NULL	
	status	varchar(100) NOT NULL	
	reg_no	varchar(100) NOT NULL	
	LOGIN_id	int(11) NOT NULL	

TABLE NAME: POLICE

	Field	Type	Comment
	id	int(11) NOT NULL	
	name	varchar(100) NOT NULL	
	phnno	varchar(100) NOT NULL	
	place	varchar(100) NOT NULL	
	district	varchar(100) NOT NULL	
	pin	varchar(100) NOT NULL	
	post	varchar(100) NOT NULL	
	status	varchar(100) NOT NULL	
	email	varchar(100) NOT NULL	
	LOGIN_id	int(11) NOT NULL	

TABLE NAME: CONTACTS

	Field	Type	Comment
	id	int(11) NOT NULL	
	name	varchar(100) NOT NULL	
	phnno	varchar(100) NOT NULL	
	POLICE_id	int(11) NOT NULL	

TABLE NAME: HELP

	Field	Type	Comment
	id	int(11) NOT NULL	
	date	varchar(100) NOT NULL	
	help	varchar(100) NOT NULL	
	time	varchar(100) NOT NULL	
	DRIVER_id	int(11) NOT NULL	
	latitude	varchar(100) NOT NULL	
	longitude	varchar(100) NOT NULL	

TABLE NAME: LOCATION

	Field	Type	Comment
	id	int(11) NOT NULL	
	latitude	varchar(100) NOT NULL	
	longitude	varchar(100) NOT NULL	
	LOGIN_id	int(11) NOT NULL	

TABLE NAME: MESSAGE

	Field	Type	Comment
	id	int(11) NOT NULL	
	date	varchar(100) NOT NULL	
	message	varchar(100) NOT NULL	
	DRIVER_id	int(11) NOT NULL	
	PARTNER_id	int(11) NOT NULL	

TABLE NAME: NOTIFICATON

	Field	Type	Comment
	id	int(11) NOT NULL	
	date	varchar(100) NOT NULL	
	notification	varchar(100) NOT NULL	

TABLE NAME: PARTNER

	Field	Type	Comment
	id	int(11) NOT NULL	
	name	varchar(100) NOT NULL	
	phnno	varchar(100) NOT NULL	
	place	varchar(100) NOT NULL	
	district	varchar(100) NOT NULL	
	pin	varchar(100) NOT NULL	
	post	varchar(100) NOT NULL	
	LOGIN_id	int(11) NOT NULL	
	DRIVER_id	int(11) NOT NULL	
	email	varchar(100) NOT NULL	
	photo	varchar(150) NOT NULL	

TABLE NAME: SETTINGS

	Field	Type	Comment
	id	int(11) NOT NULL	
	ringmode	varchar(100) NOT NULL	
	brightness	varchar(100) NOT NULL	
	speed	varchar(100) NOT NULL	
	DRIVER_id	int(11) NOT NULL	
	screenlock	varchar(100) NOT NULL	

TABLE NAME: WORK

	Field	Type	Comment
	id	int(11) NOT NULL	
	workname	varchar(100) NOT NULL	
	status	varchar(100) NOT NULL	
	AMBULANCE_id	int(11) NOT NULL	

4. SYSTEM TESTING AND IMPLEMENTATION

4.1 SYSTEM TESTING

Testing is an activity to verify that a correct system is being built and is performed with the intent of finding faults in the system. However not restricted to being performed after the development phase is complete, but this is to carry out in parallel with all stages of system development, starting with requirements specification. Testing results, once gathered and evaluated, provide a qualitative indication of software quality and reliability and serve as a basis for design modification if required. A project is said to be incomplete without proper testing.

Testing is an activity to verify that a correct system is being built and is performed with the intent of finding faults in the system. However not restricted to being performed after the development phase is complete, but this is to carry out in parallel with all stages of system development, starting with requirements specification. Testing results, once gathered and evaluated, provide a qualitative indication of software quality and reliability and serve as a basis for design modification if required. A project is said to be incomplete without proper testing.

4.2 TESTING METHEDOLOGIES

- Unit Testing
- Integration Testing
- Validation Testing
- Output Testing
- User Acceptance Testing

Unit Testing

The first level of testing is called as unit testing. Here the different modules are tested and the specification produced during design for the modules. Unit testing is essential for verification of the goal and to test the internal logic of the modules. Unit testing is conducted to different modules of the project. Errors were noted down and corrected down immediately and the program clarity was increased. The testing was carried out during the programming stage itself. In this step each module is found to be working satisfactory as regard to be expected out from the module

Integration Testing

The second level of testing includes integration testing. It is a systematic testing of constructing structure. At the same time tests are conducted to uncover errors with the interface. It need not to be the case, that software whose modules when run individually showing results will also show perfect results when run as a whole. The individual modules are tested again and the results are verified. The goal is to see if the modules integrated between the modules. This testing activity can be considered as testing the design and emphasizes on testing modules interaction.

Validation Testing

The next level of testing is validation testing. Here the entire software is tested. The reference document for this process is the requirement and the goal is to see if the software meets its requirements. The requirement document reflects and determines whether the software functions as the user expected. At culmination of integration testing, software is completely assembled as a package and corrected and a final series of software test validation test begins. The proposed system under construction has been tested by using validation testing and found to be working satisfactory. Data validation checking is done to see whether the corresponding entries made in different tables are done correctly. Proper validation checks are done in case of insertion and updating of tables, in order to see that no duplication of data has occurred. If any such case arises proper warning message will be displayed. Double configuration is done before the administrator deletes a data in order to get positive results and to see that o data have been deleted by accident

Output Testing

The output of the software should be acceptable to the system user. The output of requirement is defined during the system analysis. Testing of the software system is done against the output and the output testing was completed with success

User Acceptance Testing

An acceptance test has the objective of selling the user on the validity and reliability of the system. It verifies that the system procedures operate to system specification and the integrity of the vital data is maintained.

4.3 SYSTEM IMPLEMENTATION

System implementation is the final phase i.e., putting the utility into action. Implementation is the state in the project where theoretical design turned into working system. Implementation involves the conversion of a basic application to complete replacement with a computer system. It is the process of converting to a new or revised system design into an operational one. During the design phase, the products structure, its undergoing data structures, the general algorithms and the interfaces and control/data linkages needed to support communication among the various sub structures were established

Implementation process is simply a translation of the design abstraction into the physical realization, using the language of the target architecture. Implementation includes all those activities that take place to convert from the old system to the new. The new system may be totally new replacing an existing manual or automated system, or it may be major modification to an existing system. In either case, proper implementation is essential to provide a reliable system to meet organizational requirements. There are three types of implementations:

- Implementation of a computer system to replace a manual system.
- Implementation of a new computer system to replace an existing one.
- Implementation of a modified application to replace an existing one, using the same computer.

The common approaches for implementation are:

Parallel Conversion

In parallel conversion the existing system and new system operate simultaneously until the project team is confident that the new system is working properly. The outputs from the old system continue to be distributed until the new system has proved satisfactorily parallel conversion is a costly method because of the amount of duplication involved.

Direct Conversion

Under direct conversion method the old system is discontinued altogether and the new system becomes operational immediately. A greater risk is associated with direct conversion is no backup in the in the case of system fails.

Pilot Conversion

A pilot conversion would involve the changing over of the part of the system either in parallel or directly. Use of the variation of the two main methods is possible when part of the system can be treated as a separate entity.

User Training

After the system is implemented successfully, training of the user is one of the most important subtasks of the developer. For this purpose, user manuals are prepared and handled over to the user to operate the developed system. Thus, the users are trained to operate the developed system. In order to put new application system into use, the following activities were taken care of:

- Preparation of user and system documentation.
- Conducting user training with demo and hands on.
- Test run for some period to ensure smooth switching over the system

The major implementation procedures are:

- Test plans
- Training
- Conversion

Test Plans

The implementation of a computer- b a s e d system requires that the test data be prepared and the system and its elements be tested in a structured manner

Training

The purpose of training is to ensure that all the personal who are to be associated with the computer- b a s e d business system possesses the necessary knowledge skills. A s the system provides user friendliness only basic training is needed.

Conversion

It is the process of performing all of the operations that results directly in the turnover of the new system to the user.

Conversion has two parts: The creation of a conversionplan at the start of the development phase and the implementation of the plan throughout thedevelopment phase. The creation of a system changes over plan at the end of the developmentphase and the implementation of the plan at the beginning of the operation phase.

4.4 SYSTEM MAINTANANCE

The maintenance is an important activity in the life cycle of a software product. Maintenance includes all the activities after the installation of software that is performed to keep the system operational. The maintenance phase of a software life cycle is the time period in which a product performs useful work.

Maintenance is classified into four types:

- Corrective Maintenance
- Adaptive Maintenance
- Perfective Maintenance
- Preventive Maintenance

Corrective Maintenance

Corrective maintenance refers to changes made to repair defects in the design, coding, or implementation of the system. Corrective maintenance is often needed for repairing processing or performance failures or making changes because of previously uncorrected problems or false assumptions. Most corrective maintenance problems surface soon after the installation. When corrective maintenance problems surface, they are typically urgent and need to be resolved to curtail possible interruptions in normal business activities.

Adaptive Maintenance

Adaptive maintenance involves making changes to an information system to evolve its functionality or to migrate it to different operating environment. Adaptive maintenance is usually less urgent than corrective maintenance because of business and technical changes typically occur some period of time.

Perfective Maintenance

Perfective maintenance involves making enhancements to improve processing performance, interface usability, or to add desired, but not necessarily required, system features. Many system professionals feel that perfective maintenance is not really the maintenance but new development.

Preventive Maintenance

Preventive maintenance is the only maintenance activity which is carried out without formal maintenance request from the user. When a software company or maintenance agency realizes that the methodologies used in a program have become obsolete, it may decide to develop or modify parts of the program, which do not confirm to the current trends. Of these types, more time and money are spending on perfective than on corrective and adaptive maintenance together

CONCLUSION

The Driver Vigilance System addresses the pressing issue of distracted and drowsy driving, a leading cause of road accidents and fatalities. This system employs advanced machine learning and image processing techniques to monitor drivers in real time, ensuring a safer driving experience.

The Driver App plays a pivotal role by detecting drowsiness and distractions through smartphone cameras and sensors. It alerts drivers in critical situations, while also facilitating efficient communication with emergency services, hospitals, and law enforcement. The Partner App enhances safety further by enabling real-time monitoring and personalized settings to promote attentive driving.

The inclusion of specialized applications for police, ambulances, and hospitals ensures seamless coordination during emergencies, providing timely assistance to those in need. A dedicated website integrates data flow between stakeholders, streamlining operations and enhancing effectiveness.

This system sets a new benchmark for road safety by combining innovative technologies with practical functionalities. Through early detection of potential hazards, timely alerts, and efficient emergency responses, the Driver Vigilance System aspires to significantly reduce accidents and save lives on the road.

BIBLIOGRAPHY

Books

- "Django for Beginners" by William S. Vincent: This book provides a comprehensive guide to Django web development, covering topics such as user authentication, admin interfaces, and managing data.

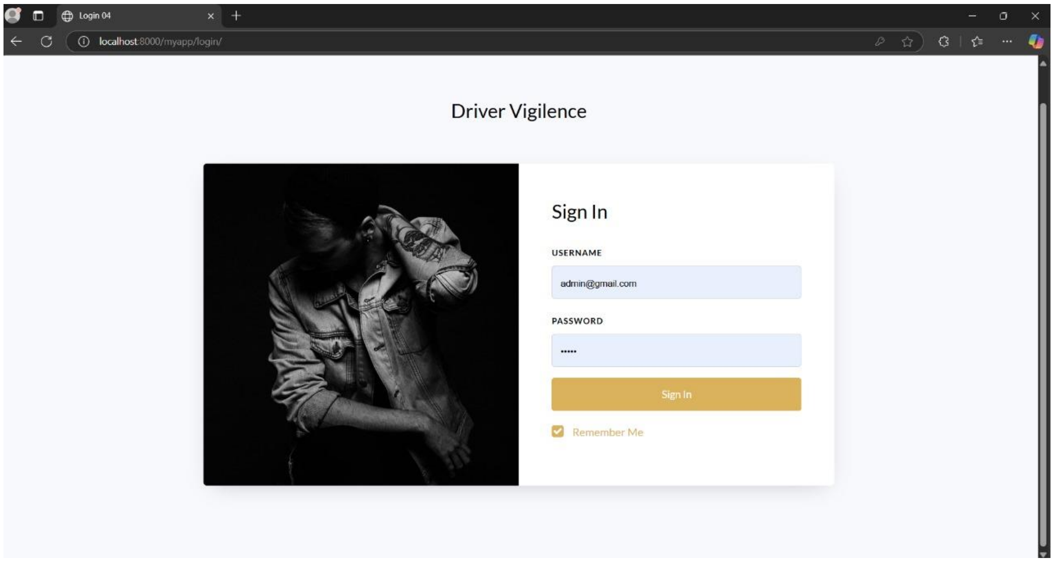
- "Python Crash Course" by Eric Matthes: While not specific to Django or web development, this book covers Python fundamentals.
- "Django for APIs: Build web APIs with Python & Django" by William S. Vincent: This book focuses on building APIs with Django.
- "Learning Django Web Development" by Sanjeev Jaiswal: This book is aimed at beginners and covers Django basics, including user authentication, views, templates, and forms.
- "Programming Android: Java Programming for The New Generation of Mobile Devices" by Zigurd Mednieks: This book provides a comprehensive guide to Android app development using Java, which would be useful for implementing the user module's mobile application.
- "SQL QuickStart Guide: The Simplified Beginner's Guide to Managing Analyzing, and Manipulating Data With SQL" by Walter Shields: This book would be beneficial for understanding database management and querying, which is essential for all modules interacting with the database.

Links

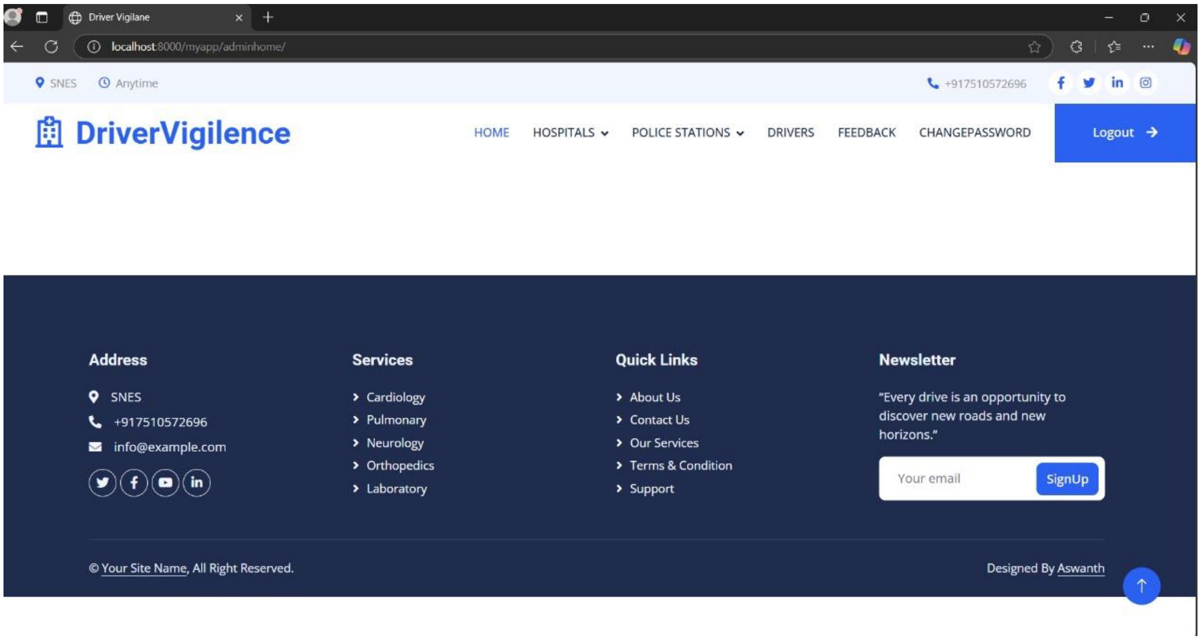
1. https://stackoverflow.com/questions/tagged/django
2. https://developer.android.com/studio
3. https://developer.android.com/training/data-storage/sqlite
4. https://scikit-learn.org/stable/documentation.html
5. https://www.django-rest-framework.org/
6. https://firebase.google.com/docs

APPENDIX 1: Screenshot

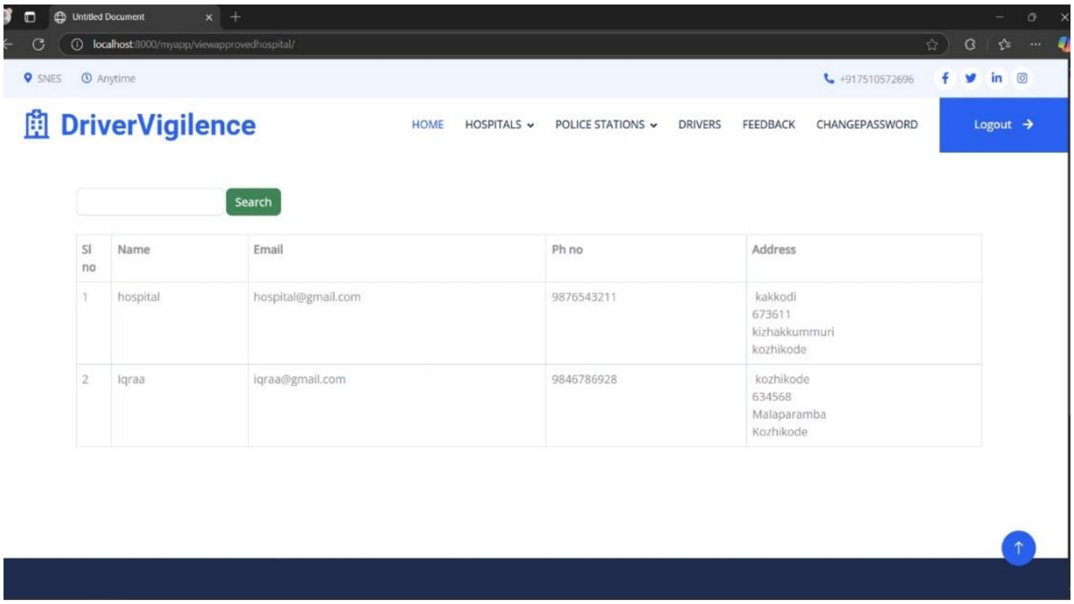
Web Login Screen



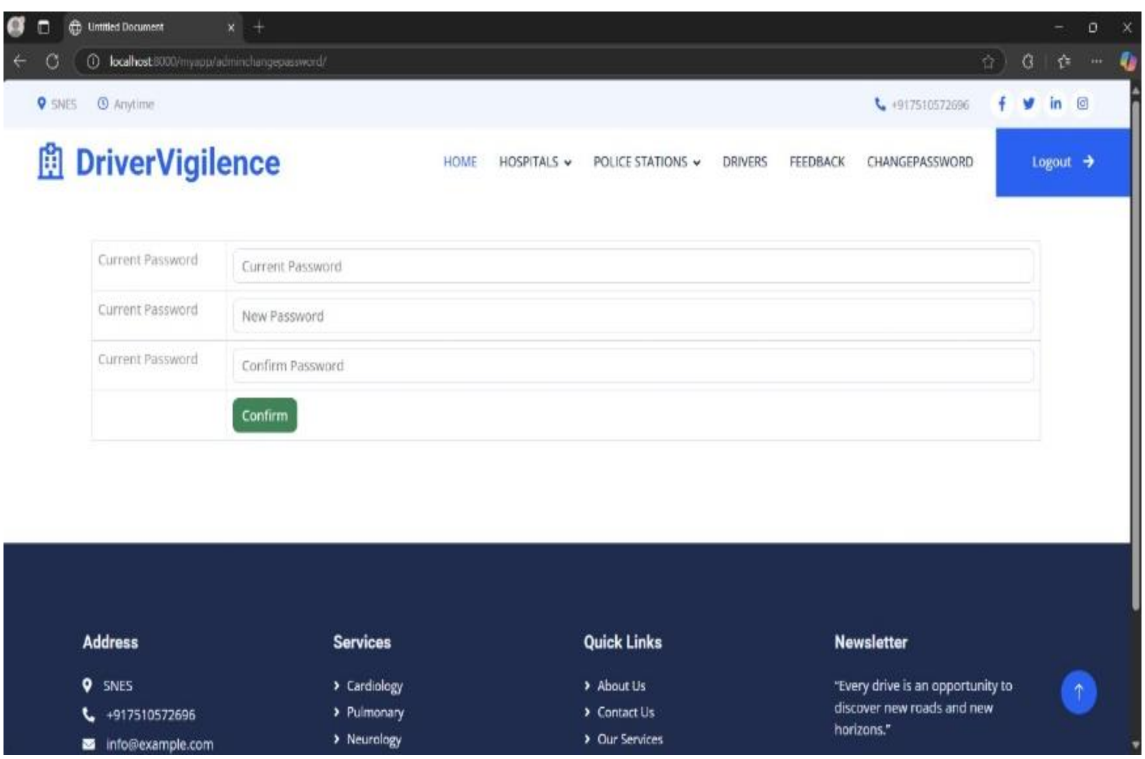
Web Admin Home Screen



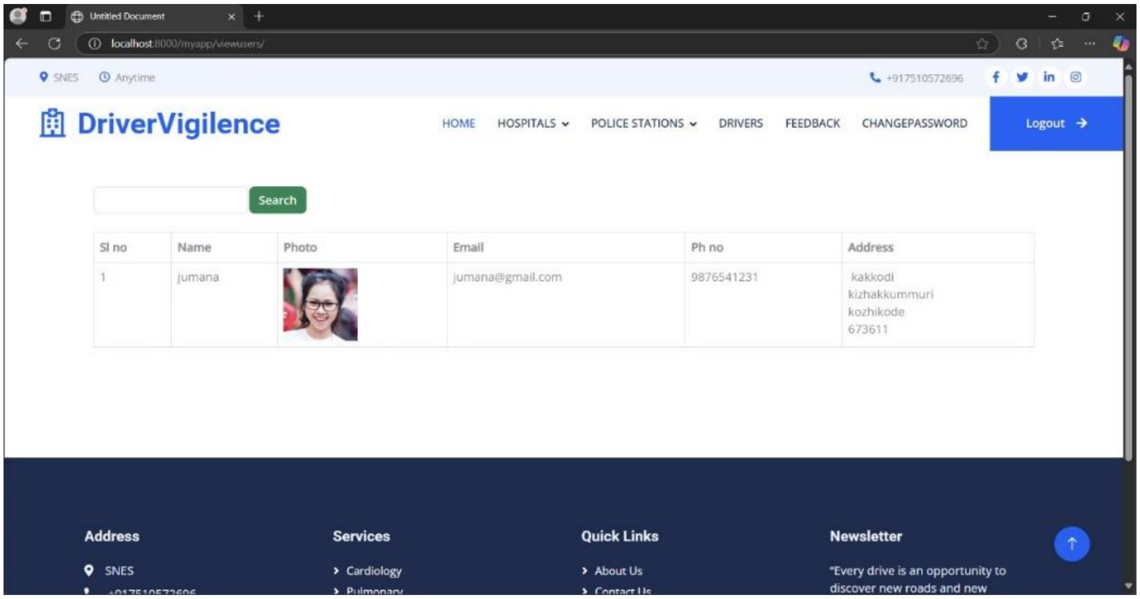
Hospital screen



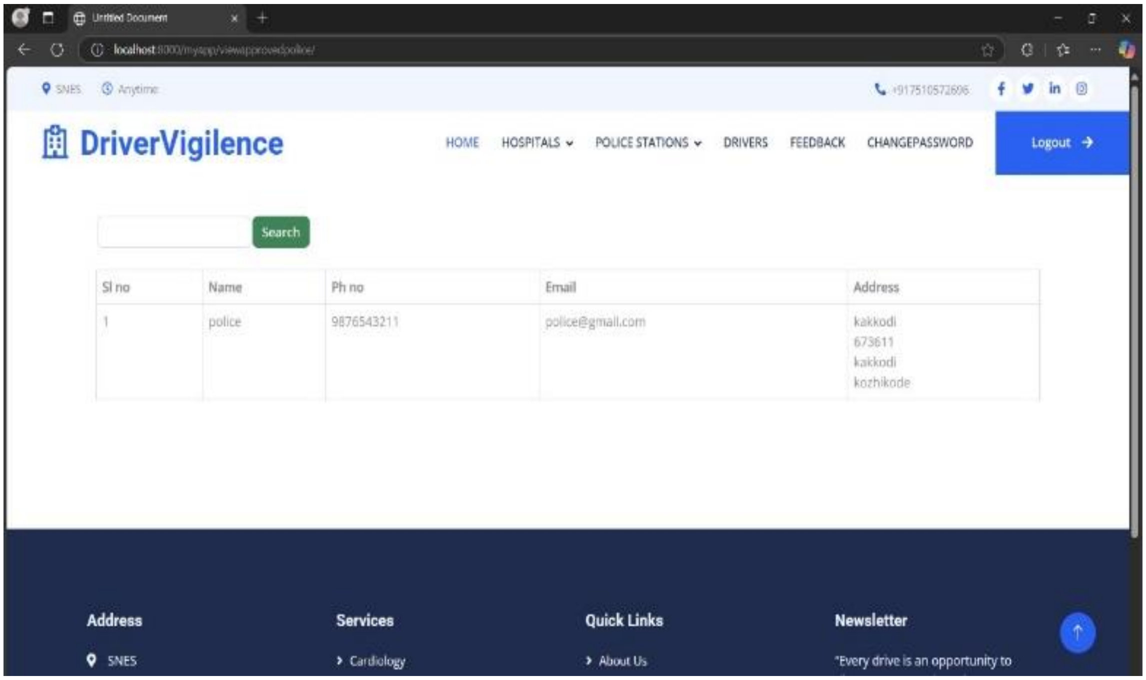
Change Password screen



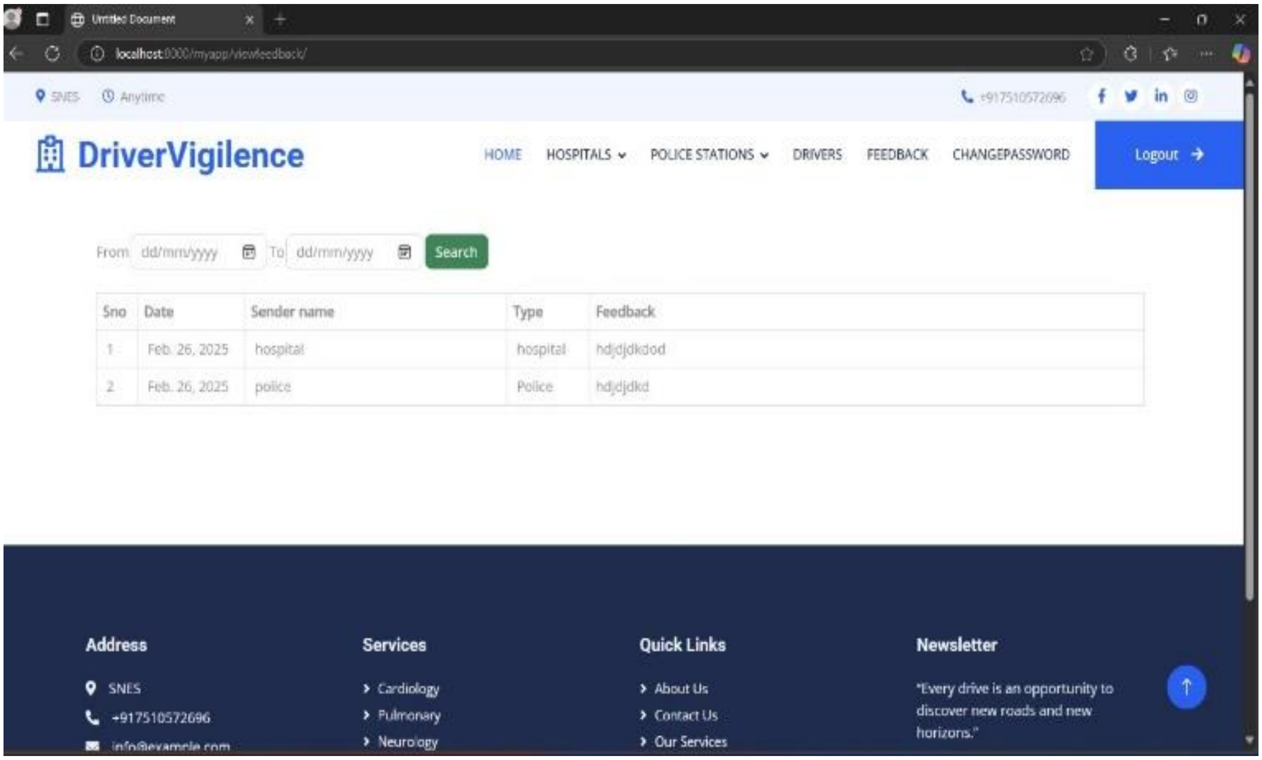
Driver screen



Police screen



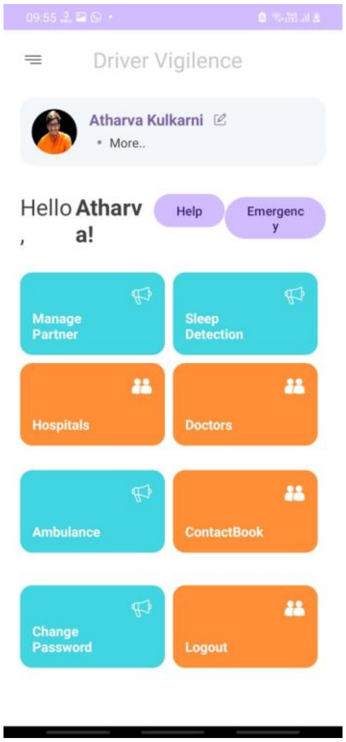
FeedBack Screen



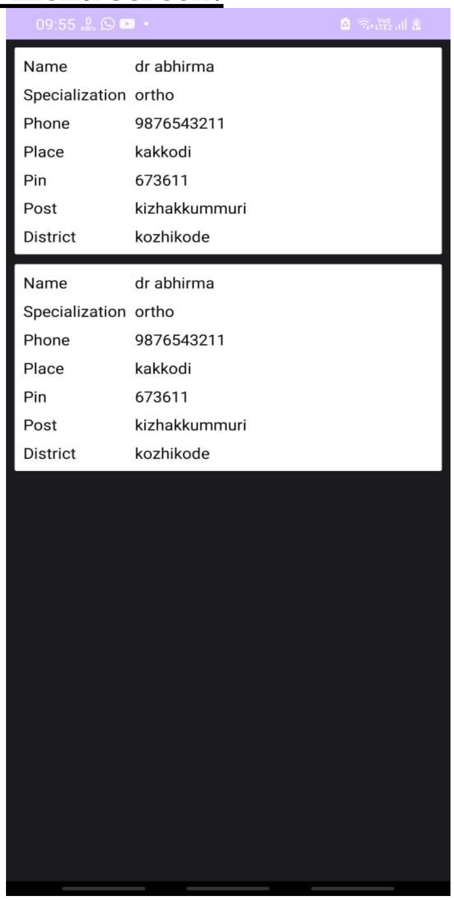
Android Screen

Ip screen:

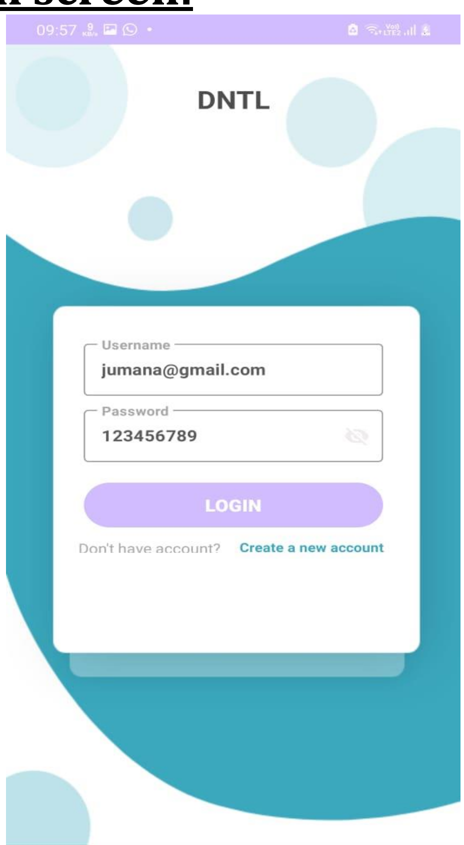
Home screen:



Home menu screen:



Login screen:



Driver screen: